

International Conference on Machine Learning and Data Engineering

DeepGuardNet: A Novel CNN Architecture for DeepFake Image Detection

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Abstract

We are in the world where information is the ubiquitous but the authentication of that information is cumbersome. Deepfake technology have achieved a tremendous growth in the digital era. Deepfake is a synthetic media audio, video or images that appear to be realistic though they are fake or fabricated. Deepfake are created by Generative AI techniques that understand the probabilistic distribution of the data. Appropriate detection systems are necessary to prevent the dissemination of misleading information and guarantee the authenticity and integrity of data. Detecting deepfake contents in this digital era is very challenging due to the realistic nature of fake images. In this paper, we present an enhanced CNN architecture, DeepGuardNet, a deepfake detection model that is simple and effective for determining whether images are real or fake. DeepGuardNet is a straightforward, sequential, and robust network designed for deepfake recognition and detection. Additionally, our network has an enhanced ability to detect tampered content with fewer parameters due to the use of separable convolution. In this work, we utilize depthwise separable convolution to efficiently extract deepfake features. The DeepGuardNet architecture effectively captures deepfake image features in both the spatial and depth dimensions. Experimental study on Celeb-DF dataset demonstrated the competence of the proposed method with an accuracy of 91% when compared with conventional methods. The proposed DeepGuardNet architecture is productive in terms of the better feature extraction and reduced computational complexity.

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering

Keywords: Deepfake; DeepGuardNet; MesoNet; Separable Convolution; Convolutional Neural Network; Deepfake Detection; Deep Learning

1. Introduction

The smartphones gained much acceptance over the past few decades and the growing popularity of digital media have made very common digital objects. This growth in the digital media contents also possessed the falsification of these contents. Digital image forensics is the area which aims to regulate the circulation of tampered data in the

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social medias. Deepfakes are synthesized media developed with the help of artificial intelligence and advanced deep learning algorithms, that fabricate highly realistic videos, images and audios.[22] Despite their numerous beneficial applications, deepfake images raise serious concerns related to security, privacy of people and about spreading misinformation. Now a days generative AI models including generative adversarial networks(GAN), variational auto encoders and stable diffusion[21] have the ability to create fake faces that are very realistic so that it is tedious for a software to detect it[3]. In the current era, Artificial Intelligence based generative models led to a scenario where the fake content generation is easier day by day. So developing suitable methodologies for the deepfake detection in social media is the need for the hour now. To tackle these social issues the deepfake detection technology had gained wide attention. Several approaches were developed for the deepfake detection. Several methods like local binary pattern recognition[4], error level detection[5], facial land marks[6]. The contribution of the work is three fold,

- DeepGuardNet is a simple, sequential and robust convolutional neural network(CNN) model and its main objective is to identify forged images and videos.
- In addition the adoption of separable convolution, the architecture aims to enhance the ability to capture the deepfake features with fewer parameters.
- DeepGuardNet illustrates better performance over Celeb-DF dataset, achieving an accuracy of 91%.

A separable convolution or depth wise separable convolution is a set of spatial convolution performed over each input channel separately followed by a 1×1 convolution, projecting the output channel onto a fresh space[7]. The feature extraction using depth wise separable convolution is extremely good at tell apart real and false images[20]. The media forensics typically involves analysing the inconsistencies and AI based approaches that use machine learning to determine real and fake contents. In the modern era of technology, the deepfake media content have revolutionized the world because of its wide popularity and there is an exigency for the development of algorithms that can accurately detect the fake videos or image content. Here in this paper we discuss about such a technique to detect the fabrication made in the digital videos, we have proposed a new deep architecture DeepGuardNet that is inspired from Meso4 developed by Afchar[1] et al. We have applied DeepGuardNet on the frames of Celeb-DF dataset and overall got better result.

This section gave brief an introduction to deepfake detection techniques. Section 2 includes the short and summarized works related to deepfake detection. Section 3 presents the proposed method and its salient features. Section 4 covers the results and inferences from the work and section 5 summarizes the work.

2. Literature Summary

2.1. Deepfake Recognition & Detection Methods

Author et al.	Year	Models for Investigation	Dataset	Remarks
Darius Afchar et al.[1]	2018	Meso4, MesoInception	Deepfake, Face2Face	Focuses on Mesoscopic Properties of image
Mohammed Thajeel Abdullah et al.[4]	2023	DenseNet121 + CNN	FFHQ, Celeb-DF	MesoNet with pre-processing module
Zhiming Xia et al.[9]	2022	MesoNet	Celeb-DF	MesoNet with pre-processing module
Nan Wu et al.[10]	2022	MSPNN	FaceForensics++, Celeb-DFv2	Simultaneously capture micro, meso and macro semantics

Rimsha Rafique et al.[5]	2023	ResNet18	Dataset by Yonsei University	It is a robust method. ELA manipulates at pixel level
Aminollah Khormali et al.[11]	2024	Self-supervised graph Transformer	FaceForensics++, Celeb-DF	The model exploited transformer based self-supervised graph strategy for deepfake detection, using contrastive learning.
Ying Xu et al.[10]	2023	Multi-Channel Xception Attention Pair-wise Interaction	FaceForensics++, Celeb-DF	It uses pairwise learning and complementary color space information.
Deressa Wodajo et al[13].	2021	CViT	DFDC	The combined learning capacity of CNNs and Transformer enables the model to correlate every pixel of an image.
Akash Saxena et al.[6]	2023	XceptionNet	DFDC	Efficient for quickly accessing deepfake films using limited computing resources.
Kurniawan Nur Ramadhani et al.[12]	2024	ViViT	Celeb-DF V2	The system uses ViViT with DSC and CBAM to analyze facial landmark features
Yue Wu et al.[19]	2019	ManTra-Net	CASIA, NIST	ManTra-Net is used for localization of forged pictures

Table 1: Summary of DeepFake Works

In digital image forensics, the conventional methods include splicing where a portion of the image is manipulated but in the current scenario entire whole fake images and videos can be generated using generative models.

In the social media era it is important to validate the authenticity and integrity of data what we are observe and share. Deepfake detection plays a vital role in dealing with false information, protecting people's identity, preventing fraudulent activities and maintaining political stability. Moreover, a comprehensive method is required to preserve the cybersecurity regulations and preventing the erosion of public trust in digital platforms. As deepfake technology continuously achieve rapid growth, the ability to differentiate between the realistic and tailored content becomes essential to safeguard the integrity of information in the digital media.

3. Proposed Method- DeepGuardNet

In this segment, we present an efficient CNN architecture DeepGuardNet that is simple and light weight for enhanced deepfake detection. This simple model has the capability to extract fake features so that the architecture becomes efficient. DeepGuardNet model is less complex, has less number of training parameters and reduced computational complexity. This network was designed by leveraging inspiration from MesoNet Proposed by Darius Afchar et al[1]. In MesoNet, the mesoscopic feature which lies in between microscopic and macroscopic properties were exam-

ined and used for the manipulation of forged images. DeepGuardNet utilizes the depth-wise separable convolutions to decrease the computational complexity, less memory usage, less number of training parameters. In addition to the MesoNet layers, the model used one more convolution layer with separable convolution instead of the traditional convolution method. As compared to the conventional methods it has less number of parameters which can facilitate to prevent over fitting, which better the generalization to that has not been seen before the features extracted[2].

The DeepGuardNet have five consecutive separable convolution layers that inturn provide were used along with batch normalization and max pooling layers. In this network the data in RGB color space is served by resizing it into the frame size of $256 \times 256 \times 3$ with three color channels, then passed into the convolutional layers. Here in separable convolution the convolution is performed in two steps: a depth wise convolution followed by pointwise convolution. Batch normalization technique is applied after each convolution for increasing its training speed and stability. For reducing the spatial dimensions of the feature map we use maxpooling layers that helps to capture the most relevant features. To enhance their reliability, the convolutional layers uses ReLU activation function to bring in non-linearities and normalize its output to avert vanishing gradient effect and the fully connected layers use Dropout to stabilize the model and prevents over fitting. DeepGuardNet is a compact network having 3204 trainable parameters out of 3332. After the feature extraction process the model classifies the result into real and fake images. Experiments are conducted in Celeb-DF dataset and obtained better and efficient result as compared to others.

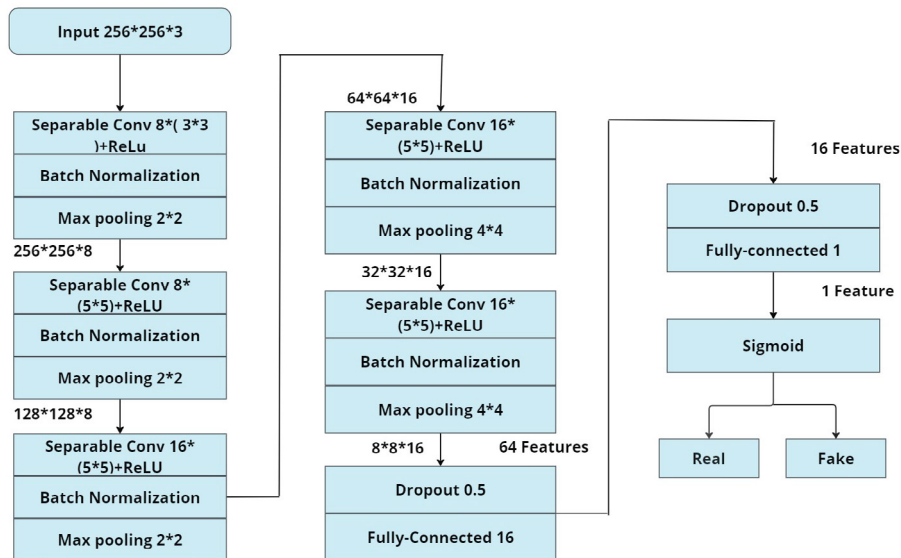


Fig. 1. DeepGuardNet Architecture

4. Result Analysis and Discussions

In the following part we will discuss about the results of the experiment along with a comprehensive discussion about the findings. The efficiency of this method is evaluated by the performance measures and then comparing it with the other existing methods.

4.1. Dataset

The dataset used in this work is Celeb-DF[18], which is a well known challenging dataset that makes a benchmark dataset for accessing deepfake detection. It contains 5639 deepfake videos and 590 real videos. These videos have average length of 13 seconds which have frame length of 30 frames per second. For training we have used 140002 and the validation set have 39428 image frame which are classified into real and forged images. The dataset has been

divided in the ratio of 8:2 for training and validation. The original videos were taken from the YouTube and the tampered videos were generated by swapping the faces[8]. Because of the high quality of the contents, Celeb-DF is one among the most challenging dataset to handle deepfake detection algorithm to handle fundamental deepfake hurdles including flickering, facial wrapping and unnatural expressions. Some of the sample images from the dataset is given in the Fig.2.

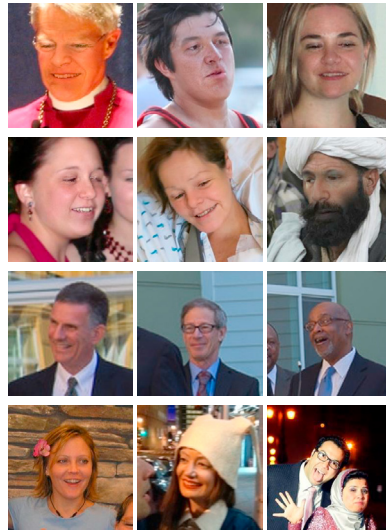


Fig. 2. Samples from Celeb-DF dataset

4.2. Experimental Setup & Hyper Parameters Used

DeepGuardNet network was implemented using python 3.10 and keras 3.4.1 module on Jupyter Notebook. The parameters used are ADAM optimizer used for the weight optimization along with default parameters ($\beta_1 = 0.9$ and $\beta_2 = 0.999$), the learning rate of 0.001, batch size of 5 and 20 epochs. This network uses Binary Cross Entropy as its loss function, with a 0.5 dropout.

Table 2. Hyper parameters used for DeepGuardNet

Parameter	Value
No. of epoch	20
Batch size	5
Optimizer	ADAM
Loss function	Binary Cross Entropy
Learning rate	0.001
Dropout	0.5
Parameter β_1	0.9
Parameter β_2	0.999

4.3. Computing Metrics

Evaluation parameters are utilized in machine learning to quantify the success rate of a model. Confusion metrics are among the most commonly used evaluation techniques; they are employed to calculate the performance efficiency of the test data by contrasting the actual and predicted data[5]. These metrics have four values: Correct Positive (CP), Correct Negative (CN), Incorrect Positive (IP), and Incorrect Negative (IN). This framework can be evaluated using Accuracy, Precision, Recall, and F1-Score, calculated using the formulas shown in Equations (1) to (4):

$$Recall = \frac{CP}{CP + IN} \tag{1}$$

$$Precision = \frac{CP}{CP + IP} \tag{2}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3}$$

$$Accuracy = \frac{CP + CN}{CP + CN + IP + IN} \tag{4}$$

By using the Celeb-DF dataset on our model, we achieved an accuracy of 91%. The learning curves below provide detailed insights into the performance. These curves, shown in Fig(3) to Fig(5), illustrate the training and validation accuracy over a set of training data, demonstrating a steady improvement in performance.

Comparing with other models, our DeepGuardNet achieved better outcome on the Celeb-DF dataset. Table 3 illustrates the class-wise precision, recall, f1-score on the Celeb-DF dataset and Table 4 shows a collation between DeepGuardNet and other models over Celeb-DF dataset.

Table 3. Model Performance Metrics

Model	Class	Prec	Rec	F1
MesoNet	Fake	87%	52%	65%
	Real	66%	92%	77%
DeepGuardNet	Fake	92%	88%	90%
	Real	89%	93%	91%

Table 4. Comparison over various existing models

Author et al.	Method	Accuracy
Yuezun Li[15]	Xception-c40	65.5%
Elhassan et al.[17]	DFT-MF	71.29%
Afchar et al.[1]	MesoNet	72%
Ahmad et al.[16]	ResViT	80.48%
	DeepGuardNet(Proposed)	91%

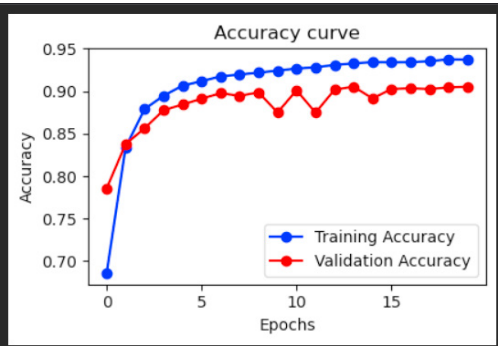


Fig. 3. DeepGuardNet Accuracy Curve

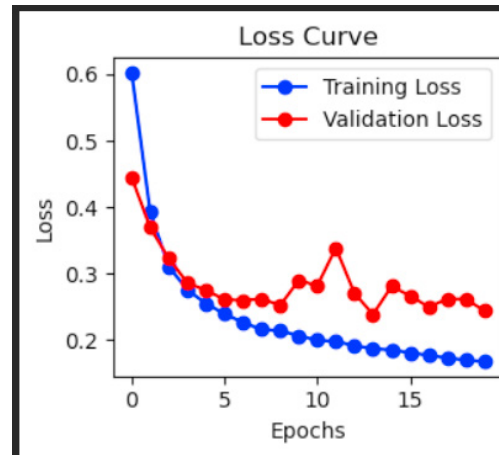


Fig. 4. DeepGuardNet Loss Curve

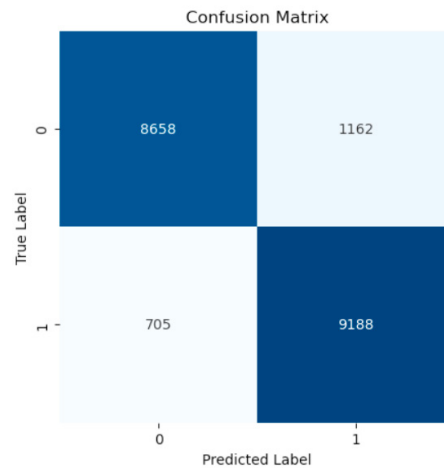


Fig. 5. DeepGuardNet Confusion Matrix

5. Conclusion

In the digital age, it is crucial to keep the integrity of data what we see and share in digital media. In this research, we have developed an model DeepGuardNet for addressing the evolving challenge possessed by the AI generated false contents in the digital media, that reduces the wide spread of the misinformation and safe guard the probity of online media. We have used the mesoscopic features of an image for the detection with the help of separable convolution. Using this method, results demonstrate that separable convolution is superior to the conventional method because of its better performance. Separable convolution maintains or even improves the model with lesser number of parameters and computational complexity. As comparing with the traditional models, we can state that separable convolution can create a better impact on the forgery detection. Experimental results showcased a detection rate of 91% accuracy on image frames of Celeb-DF dataset. The proposed model is efficient in identifying real and fake data that shows its potential in realistic implementation in safeguarding the integrity of digital contents. Overall the use of separable convolution has been found to be an imperative improvement, making a deepfake detection system more effective and reliable. This work imposes a groundwork for the future events in deepfake detection resulting in further development in this area and applied to variety of datasets and parameters.

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